

Volume 8 – Machine Learning & Predictive Intelligence Design

AI Quant Trading Platform

Version: 1.0

Document Type: Machine Learning Architecture

Classification: Advanced Intelligence Layer

1. Purpose

The Machine Learning (ML) subsystem enhances the platform by discovering patterns in historical market behavior and strategy outcomes. Unlike the rule-based AI Decision Engine, the ML subsystem focuses on prediction, evaluation, and continuous improvement using validated historical data.

The ML subsystem shall **support** trading decisions. It shall not bypass the Strategy Engine or Risk Engine.

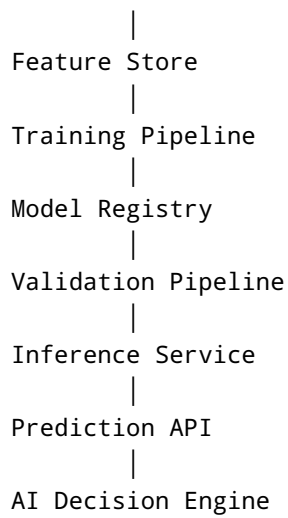
2. Objectives

The ML subsystem shall:

- Learn from historical trade outcomes
 - Rank trading opportunities
 - Predict probability of success
 - Classify market regimes
 - Detect anomalies
 - Improve recommendation quality
 - Monitor model performance
 - Support research and experimentation
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3. ML Architecture

```
graph TD;
    A[Market Data] --> B[Historical Database];
    B --> C[Feature Engineering];
```



4. Data Sources

Market Data

- OHLCV
- Tick Data
- Order Book
- Volume
- Spread
- Funding Rate
- Open Interest
- Liquidation Events

Technical Indicators

- EMA
- SMA
- RSI
- MACD
- ATR
- ADX
- VWAP
- SuperTrend
- Bollinger Bands
- Volume Profile
- MFI
- OBV

Trade History

- Entry Price

- Exit Price
 - Stop Loss
 - Take Profit
 - Holding Time
 - Position Size
 - Strategy Used
 - Trade Outcome
-

Portfolio Data

- Balance
 - Drawdown
 - Exposure
 - Allocation
-

Context Data

- BTC Trend
 - ETH Trend
 - Stablecoin Dominance
 - Session Timing
 - Market Regime
 - Exchange Status
-

5. Feature Engineering

Features include:

Price Features

- Returns
- Volatility
- Momentum
- Trend Strength

Indicator Features

- RSI
- EMA Distance
- MACD Histogram
- ATR
- ADX

Market Features

- Liquidity
- Spread

- Volume Spike
- Order Book Imbalance

Strategy Features

- Active Strategy
- Timeframe
- Risk Profile

Portfolio Features

- Exposure
 - Current Drawdown
 - Open Positions
-

6. Feature Store

The Feature Store maintains standardized inputs for all models.

Requirements:

- Versioned features
 - Timestamped records
 - Reusable across models
 - Validation before training
-

7. Model Categories

Classification Models

Purpose:

Predict whether a setup satisfies predefined quality criteria.

Regression Models

Purpose:

Estimate expected movement or outcome-related metrics.

Ranking Models

Purpose:

Rank candidate opportunities.

Anomaly Detection

Purpose:

Detect abnormal market behavior.

Market Regime Models

Purpose:

Classify market state:

- Bull
- Bear
- Range
- High Volatility
- Low Volatility

8. Candidate Algorithms

Traditional

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM

Deep Learning

- LSTM
- Transformer-based Time Series
- Temporal Convolutional Networks (future)

Experimental

- Reinforcement Learning (research only)
- Ensemble Models

Algorithm selection shall be evidence-based using validation metrics.

9. Model Training Pipeline

Data Collection

↓

Cleaning



Feature Engineering



Training



Validation



Performance Evaluation



Approval



Deployment



Monitoring

Only validated models may enter production.

10. Training Schedule

Options:

- Manual
- Weekly
- Monthly
- Triggered by sufficient new data

Production retraining requires validation and approval.

11. Validation

Metrics

Classification

- Precision
- Recall
- F1 Score
- ROC-AUC

Regression

- MAE
- RMSE

Ranking

- NDCG
- MAP

Operational

- Inference Latency
 - Stability
 - Drift
-

12. Model Registry

Every deployed model stores:

- Model ID
 - Version
 - Training Date
 - Dataset Version
 - Hyperparameters
 - Performance Metrics
 - Approval Status
 - Rollback Version
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13. Inference Service

Responsibilities

- Load active model
- Validate input features

- Produce prediction
- Return confidence
- Record inference logs

The service must remain stateless for horizontal scaling.

14. Drift Detection

Monitor:

- Feature Drift
- Data Drift
- Prediction Drift
- Performance Drift

When thresholds are exceeded:

- Alert administrators
 - Flag model for review
 - Prevent automatic promotion
-

15. Explainability

Each prediction should provide supporting factors where feasible.

Example:

- Strong higher-timeframe trend
- Elevated volume
- Low spread
- Momentum confirmation

Explainability improves transparency and user trust.

16. Online Learning

Direct online modification of production models is disabled by default.

Instead:

- Collect feedback
- Store new observations
- Evaluate offline
- Retrain under controlled conditions
- Promote only after validation

17. Reinforcement Learning (Research)

Future research module.

Potential uses:

- Position management
- Exit optimization
- Strategy parameter tuning

Constraints:

- Isolated research environment
 - Never directly controls live trading
 - Requires extensive validation
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18. Security

- Model artifacts are signed
 - Feature validation before inference
 - Dataset access controlled
 - Audit logs for model deployment
 - Version rollback supported
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19. Monitoring

Monitor:

- Prediction latency
 - Model accuracy trends
 - Drift indicators
 - Error rates
 - Feature availability
 - Resource usage
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20. Failure Handling

If the ML subsystem is unavailable:

- Continue operating using the rule-based AI Decision Engine
- Record incident
- Notify administrators

- Prevent incomplete predictions from influencing decisions
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21. Design Principles

- Models assist; they do not override risk controls.
- Production models require validation before deployment.
- Every prediction is traceable to a model version.
- Retraining is controlled and auditable.
- System behavior remains deterministic when ML is unavailable.
- Safety has higher priority than prediction accuracy.